SATYA BONDA

Principal AI Engineer · Agentic AI Architect — Multi-Agent Orchestration · MCP · RAG · LLM Evaluation · AI Governance · Azure & AWS
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SUMMARY

Principal AI Engineer and agentic AI architect with 23 years in enterprise technology: 18 in delivery leadership at IBM, Oracle, Volvo, Trane and Wipro, hands-on AI engineering since 2021 and production agentic systems since 2024. Designs and ships large language model (LLM) applications end to end: multi-agent orchestration on MCP, retrieval-augmented generation (RAG) grounded in the customer's own documents, guardrails and AI governance written in code, and evaluation, observability and cost control that gate every release. Leads cross-functional teams taking business workflows from the first workshop to scalable, auditable production, and mentors engineers and leaders. Anthropic Claude Certified Architect – Professional and Microsoft Agentic AI Solutions Architect – Expert, both applied in delivered systems.

CORE COMPETENCIES

AI solution architecture · agentic AI & multi-agent orchestration (LangGraph, OpenAI Agents SDK, CrewAI, AutoGen) · Model Context Protocol (MCP) · context engineering & memory · retrieval-augmented generation (RAG) & hybrid search · LLM evaluation & LLM-as-a-Judge · guardrails, security & AI governance · observability & cost control · Azure AI Foundry · Amazon Bedrock · LLMOps & CI/CD · machine learning · technical leadership, mentoring & reusable patterns.

SKILLS

Agentic AI & Multi-Agent Orchestration: supervisor–worker, hub-and-spoke and handoff patterns, planner–critic–executor, generator–critic, fan-out/fan-in, ReAct, tool calling, human-in-the-loop, checkpointing, durable state, model routing, verifier gates, typed handoffs between agents.

Frameworks & Protocols: Model Context Protocol (MCP, FastMCP, official SDKs), A2A, LangGraph, LangChain, OpenAI Agents SDK, Claude Agent SDK, CrewAI, AutoGen, PydanticAI, Semantic Kernel, Microsoft Agent Framework, LiteLLM, FastAPI, Streamlit, TypeScript, React, Node.js.

Claude Architecture (CCAR-P, applied): agentic solution design, model selection and tiering, context engineering, prompt caching, structured output via tool use, MCP integration, evaluation and optimization, governance and safety; Claude Code with CLAUDE.md, hooks, skills and subagents.

Microsoft Agentic AI (AB-100 · AI-103 · TITAN · Agentic Hackathon): Azure AI Foundry (Foundry Models, Agent Service, Foundry IQ knowledge bases, Bing grounding), Azure AI Search, Azure OpenAI, Copilot Studio, Microsoft 365 Copilot agents, Dynamics 365 F&O, Power Platform, Dataverse, Agent 365 governance.

Generative AI & Retrieval: RAG, GraphRAG, agentic RAG, hybrid and semantic search (BM25 + vector), reranking, embeddings, chunking, vector databases (pgvector, Pinecone, ChromaDB, FAISS, Azure AI Search, S3 Vectors), multimodal RAG, document summarization, prompt engineering, fine-tuning (SFT, LoRA, QLoRA, DPO).

Machine Learning: ML fundamentals and statistical learning; supervised learning — linear regression (polynomial, Ridge, Lasso, Elastic Net), logistic regression, k-nearest neighbors, support vector machines, decision trees, random forests, gradient boosting (AdaBoost, XGBoost), Naive Bayes; unsupervised — k-means, hierarchical clustering, DBSCAN, PCA; feature engineering, cross-validation, grid search, model evaluation (precision, recall, F1, ROC-AUC, RMSE), NLP, model deployment as APIs.

Data & Python: Python, SQL, PostgreSQL, pandas, NumPy, scikit-learn, PyTorch, TensorFlow, Transformers, Matplotlib, Seaborn, Plotly; ETL and data pipelines, Snowflake and Boomi integrations, REST APIs, JSON.

Evaluation, Observability & Reliability: golden datasets as release gates, LLM-as-a-Judge calibrated to human review, adversarial suites and red teaming, RAGAS, DeepEval, LangSmith, OpenTelemetry, precision/recall and hit@k metrics, drift detection, canary releases with automatic rollback, failover and retries.

LLMOps / MLOps & Cloud: CI/CD for models and APIs, GitHub Actions, Docker, Kubernetes, Terraform/Bicep, AWS CDK, MLflow, Azure Container Apps, Key Vault, Entra ID, Amazon Bedrock, AgentCore, ECS Fargate, Lambda, scalable serving.

AI Security, Responsible AI & Governance: least-privilege tool access, agent identity and role-based access, prompt-injection defense, data isolation, PII/PHI redaction, Azure AI Content Safety, OWASP LLM Top 10, EU AI Act, audit trails and chain of custody, cost governance.

AI Product & Leadership: AI strategy and use-case qualification, AI product management, technical leadership and mentoring, cross-functional delivery, best practices and reference architectures, executive and stakeholder communication, business outcomes and ROI, enablement and training.

CERTIFICATIONS

Anthropic Claude Certified Architect – Professional (CCAR-P), 2026 · Microsoft Agentic AI Business Solutions Architect – Expert (AB-100, 2026) · Microsoft Azure AI Apps & Agents Developer Associate (AI-103, 2026) · Microsoft TITAN Frontier Transformation Engineer (2026).

Also: PMP, PMI-RMP, CSM, SAFe POPM, ITIL V3, Azure Solutions Architect Expert, AWS Cloud Practitioner, Oracle GenAI Professional, Oracle ML Associate.

EXPERIENCE

Principal AI Engineer

Independent AI Consultant · Bengaluru, India · Mar 2026 – Present

Forward-deployed engagements: designs, builds and runs agentic systems with client teams, from the first workshop to production handover.

- **Microsoft Agentic AI Hackathon (Oct 2026, via client Fusion5):** four Azure AI Foundry builds, all scored 100/100: models deployed from code, a Foundry IQ knowledge agent, a multi-tool MCP agent with a PII guardrail, and multimodal RAG.
- **AI enablement:** Aug–Sep 2026 cohorts for EY and PwC teams, staged for developers, managers and senior leaders (20+ participants, 4.7/5 feedback).
- **Agentic CoLearn:** a curriculum-grounded tutor for Indian school students: lessons are generated only from official NCERT text and shown only after an adversarial verifier fails to refute them; mastery tracked with Bayesian Knowledge Tracing, IRT and FSRS.
- **Copilot adoption:** built the programme hub for a Microsoft 365 Copilot rollout: release train, project plan, action tracker and service catalog.

Autonomous Agentic AI Auditor — Microsoft AI Audit Packs (Claude · MCP · Multi-Agent) · Oct 2026

- **Problem:** a Microsoft partner must prove, engagement by engagement, that each AI project followed a disciplined lifecycle: up to 74 audit templates per project, written by hand from scattered statements of work, designs, transcripts and sign-offs.
- **Built:** nine agents in a fixed chain under a deterministic supervisor. They index the project's own documents (one isolated index per customer), learn each template, draft only from retrieved evidence with verbatim citations, tighten, and pass a critic on a different model and a code check of every quotation, then write back into the original Word template. Missing evidence becomes a ready-to-send client request, never invented text, and a practice library carries approved sections and critic lessons into the next engagement.
- **Outcome:** in its first engagement, an Australian public-sector AI assistant on Azure AI Foundry, it drafted all 46 required templates from 17 documents in about four unattended hours, with zero fabricated citations, 138 precise client requests and under US$10 in model cost.

Flick — Invoice Reconciliation (LangGraph · MCP · Claude) · Apr – May 2026 · Case study

- **Problem:** finance analysts at a pest-control multinational spent 30–90 minutes on each supplier invoice, proving every line against contracts, purchase orders and meter data.
- **Built:** a LangGraph manager passes each invoice to specialist agents (parser, matcher, anomaly detector, reconciler, reporter), each reaching its data through its own least-privilege MCP server. A critic on a second model vendor rejects any claim without a source, and confidence decides the route: green posts, amber goes to an analyst, red drafts a dispute that a person releases.
- **Outcome:** analysts ask in Teams instead of opening invoices one by one. In pilot it reconciled 1,200 invoices a month, 87% on first pass, in about 4 minutes against 47 by hand; a 200-invoice set labelled by the analysts tests every change overnight before a 5% canary release.

Compass — Live Dynamics 365 Agent (Azure AI Foundry · MCP) · Jun 2026 – Present · Case study

- **Problem:** staff needed answers on orders, stock, balances and store procedures without learning Dynamics 365 screens or waiting for an analyst.
- **Built:** a planner limited to 39 read-only data entities plus search over the company's own procedures; every figure computed in code, and no answer ever exceeds the user's own Dynamics role. Delivered in Teams, on the web and as MCP tool packs.
- **Outcome:** cited, company-scoped answers in production. An 88-case test battery and an LLM judge that agrees with human reviewers 95% of the time re-test it every night and open a GitHub issue when anything fails.

Quote Intelligence — Quote Comparison (Agents SDK · MCP) · Apr 2026 · Case study · Repo

- **Problem:** a homebuilder's sales consultants spent half a day mapping a competitor's quote against their own, and a generic chatbot invents plausible prices, which is fatal in a half-million-dollar decision.
- **Built:** six agents: two quote readers in parallel, alignment across the builders' different wording, an ambiguity flagger, and a verifier on a different model that re-reads both PDFs. Every price must carry its page and exact source text, plain code checks that text against the page, and all arithmetic runs in exact decimals in code.
- **Outcome:** a source-traced, cent-exact comparison in 6–8 minutes with Word and Excel outputs. On a real job it flagged 26 items for review and refused to show a headline saving until they were resolved; 63 automated checks, 26 of them adversarial, have never let a fabricated figure through.

Enterprise AI Playground — Governed Multi-Model Agent Platform, built solo (FastAPI · LiteLLM · LangGraph · Azure) · Mar 2026 · Live · Repo

- **Problem:** every team wants frontier models and agents, without unmetered spend or ungoverned access.
- **Built:** one gateway in front of 30 models from 11 providers: identity, role, budget and key checks before each call and a cost-ledger row after it; a router sends each request to an Economy, Workhorse or Premium model; agents built from 175 reusable blueprints pause for human review and climb a five-level evaluation ladder before reaching production.

AI Program Manager / Agentic AI Architect

Insight Global · Bengaluru, India · May 2024 – Feb 2026

- **Fortune-100 AI programs:** led an 8-person GenAI team across four AI programs for a Credit & Collections organization (payment insights, a governed bot store, an agentic co-worker), from use-case qualification to adoption: accounts under AI doubled, 71% utilization, $220M collected.
- **SOX 404 agentic co-worker:** control testers re-gathered and re-checked evidence by hand every period. Built a co-worker that runs a validation plan written once per control, identically each period, and ships no conclusion without cited evidence, gated by golden-set evaluations and SME review; taken from prototype to MVP and proven on three controls (open-source edition: IQR).
- **Agentic systems, MCP era:** SDLC agents that turn a brief into requirements, designs, test plans and cutover documents (CrewAI, MCP); a sales-enablement platform exposing CRM, analytics and RAG as four MCP servers; PMO and supply-chain assistants routing questions to specialist agents over deterministic calculators (OpenAI Agents SDK); an email agent with human approval and preference memory (LangGraph); a self-correcting agentic RAG on Amazon Bedrock that grades its own retrieval (hit@5 1.00).
- **Healthcare agents:** a six-agent voice front desk for dental clinics (Twilio, ElevenLabs, Deepgram); a clinical scribe producing transcript-grounded SOAP notes with billing codes (102 tests); prior-authorization automation with regex-validated CPT/ICD-10 extraction and appeal drafting.
- **Enablement:** agentic AI cohorts at EY for practitioners and leaders.

Source code and live demos: [github.com/git-bonda108 (30+ repositories)]

AI Engineer & Project Manager

Wipro · Bengaluru, India · Jul 2021 – Apr 2024

Moved from delivery leadership into hands-on AI engineering: machine learning, MLOps and generative AI applications built at Wipro and applied inside client programmes.

- **Machine learning:** built supervised and unsupervised models end to end: regression (linear, polynomial, Ridge, Lasso, Elastic Net), classification (logistic regression, k-nearest neighbors, SVM, decision trees, random forests, gradient boosting with AdaBoost and XGBoost, Naive Bayes), clustering (k-means, hierarchical, DBSCAN) and PCA, with feature engineering, cross-validation and grid search. In use: an Azure ML random-forest order classifier (85%+ accuracy) and invoice anomaly detection whose XGBoost model learns from reviewer feedback (F1 0.72 → 0.91).
- **MLOps:** model persistence and serving as REST APIs, MLflow tracking and model registry, Azure ML pipelines and batch endpoints, CI/CD for models, and drift monitoring.
- **Generative AI experiments:** a conversational RAG assistant over enterprise documents with source attribution (LangChain, FAISS, GPT-4); a natural-language-to-SQL engine (Azure OpenAI, Hugging Face, Streamlit); EN→NL fine-tuning comparing a full MarianMT fine-tune with BLOOM LoRA (BLEU, chrF, COMET); first multi-agent prototypes (LangChain, AutoGen) for SDLC documents.
- **Microsoft copilots:** Copilot Studio agents over Dataverse, Power Apps, Power Automate and SharePoint; a ServiceNow virtual agent with knowledge grounding; Copilot Studio test-case generation. AI Hackathon 2024 winner (GPT-4 brand-voice compliance).
- **ML and GenAI training:** ran long-form programmes alongside delivery: Bristlecone (6 months: ML, deep learning, LLMs, maths and statistics), National Grid (12 months: data analytics, SQL, Power BI) and Navikenz (5 months: advanced GenAI, LangChain, AutoGen, FastAPI), plus AI tracks for leaders.
- **Delivery leadership:** owned the Oracle CPQ/MDM roadmap; deployed Oracle CPQ integrated with Salesforce; MDM modernization (Riversand/Syndigo) saved ~$900K a year and 3 FTE; led the Agile team, directed the Oracle EBS and Primavera migration to OCI, and built Python pipelines into Dynamics 365.

EARLIER EXPERIENCE

Carrier (short-term contract) | Project Manager · Feb – May 2024

Oracle EBS separation: Kinaxis and LeanDNA integrations across Boomi, Oracle EBS and Snowflake.

Trane Technologies / Ingersoll Rand | Project Manager & Agile Coach · Jun 2015 – Jul 2021

Delivered the Oracle 19c upgrade across five workstreams, 10 production databases and 8 applications, under budget for the 2020 operating plan; led a zero-disruption ERP separation through the company split (President Award); integration lead with partner vendors; revenue-backlog reporting rated 4.8/5 by stakeholders; portfolio-management SME running phase-gate reviews with steering committees; Agile coach.

Volvo | IT Project Manager · 2013 – 2015 · Oracle | Project Management Consultant · 2011 – 2013

Volvo: ITSM processes and tooling, service reliability +35%. Oracle: high-tech and oil-and-gas portfolio covering bids, SOWs, budgets, forecasts and invoice reconciliation.

IBM | Project / Operations / Transition / Quality · Jan 2007 – Apr 2011

Operations Manager for five helpdesk accounts (55 direct reports); stood up 12–20 FTE service desks for four clients; data-center migration PM.

Genpact | Process Leader · Dec 2005 – Jan 2007 · HP | Quality Specialist · Jan 2004 – Nov 2005

Led global technical-support teams against SLAs; built HP's partner QA certification process.

EDUCATION

MCA, Periyar University (2003) · B.Sc. Physics (Hons.), University of Calcutta (1999)